

Transferable Class-Correction Refinement from Multiple Rotational Data Sets

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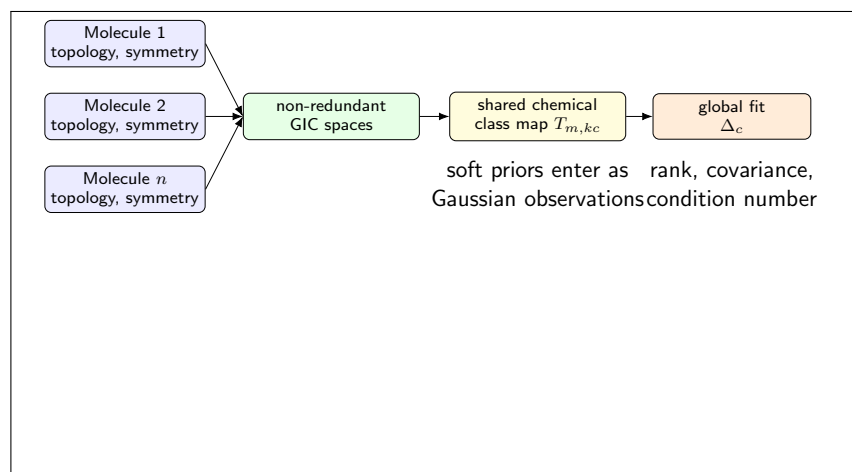
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Abstract

We introduce a multi-molecule semiexperimental refinement strategy that first validates transferable coordinate-class corrections on systems for which independent isotopologue-based refinements are available, and then applies the same model to data-poor targets. With a single molecule represented only by the parent rotational constants, or by one isotopic species, the refinement must collapse to at most one effective structural class; it cannot distinguish, for example, C–C and C–O corrections in a statistically meaningful way. Several chemically related molecules can increase the number of admissible classes because the same correction variables are observed through different molecular topologies and inertia tensors. Each molecule keeps its own topology, vibrational corrections, and non-redundant coordinate basis, while the fitted variables are common corrections for chemically defined classes such as short C–C, long C–C, carbonyl C–O, and single C–O. Maleic, phthalic, and succinic anhydrides, together with glycine conformers I and II, provide validation cases because multiple related structures and isotopologue data are available. Two deployment routes then follow: a one-species anhydride target can be integrated with the validated anhydride family, and the same multistructure machinery can select semiexperimental reference molecules most similar to the local families in a new target, recompute them at the same level, and transfer the resulting class corrections.

TOC Graphic



Introduction

Semiexperimental structure refinement is usually formulated molecule by molecule. This is natural when isotopic substitution is rich enough to determine an independent structure, and such data provide the most direct way to validate a new refinement model. The intended application, however, is broader: many larger or related molecules may have only the parent rotational constants, or a single isotopic species. In that limit a molecule-by-molecule refinement can support only a very small number of effective classes, typically a single global correction. A homologous family can carry more information because the same fitted class corrections are projected through several independent molecular graphs. A data-poor anhydride, for example, should not be refined as an isolated one-species problem; it can be embedded in the validated anhydride family so that the other members supply the complementary information needed to separate heavy-atom classes. A second route is reference-library driven: for each local family present in a new molecule, the most similar molecules are selected from a curated semiexperimental accurate-geometry library, recomputed at the same electronic-structure level used for the target, and used to estimate corrections to the target parameters.

Here we formulate this idea as a global refinement problem. For molecule m and generated internal coordinate q_k ,

$$q_{m,k}^{\text{SE}} = q_{m,k}^{\text{QC}} + T_{m,kc} \Delta_c, \tag{1}$$

where $T_{m,kc}$ maps the generated coordinate onto class c and Δ_c is shared across all molecules. The mapping is coefficient based: if a symmetry-adapted coordinate contains several primitive stretches, bends, torsions, or out-of-plane terms, the class correction is projected through the primitive coefficients rather than assigned by a binary label.

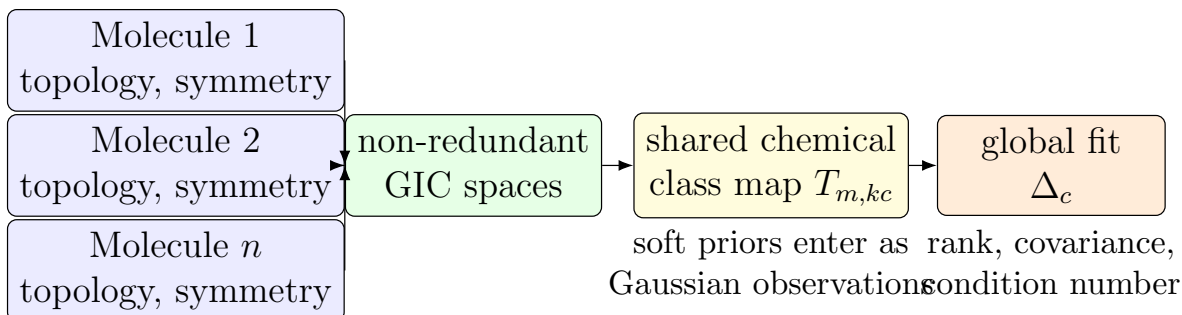


Figure 1: Multi-molecule class-correction refinement. Each molecule contributes its own topology-derived non-redundant coordinate space; chemically defined class maps connect these local spaces to a shared set of global corrections.

Model

The coordinate classes are type aware. Stretches use unordered atom pairs and may be refined by value windows, which separates, for example, carbonyl and single C–O corrections without introducing molecule-specific rules. Bends preserve the central atom. Torsions are classified by the two central atoms, i.e., by the central bond. Out-of-plane coordinates are classified by their central atom. When element symbols are too coarse, atomic synthons provide a continuous local-environment descriptor. In the present implementation this information is condensed into an effective atomic number, Z_{eff} , and two atoms are assigned to the same class when their Z_{eff} values differ by less than a chosen threshold. This prevents chemically different relations from being merged by a purely textual or fully sorted atom list.

Soft priors are ordinary Gaussian observations on the class correction,

$$\Delta_c = \Delta_c^0 \pm \sigma_c, \quad (2)$$

and enter the same weighted least-squares problem as the spectroscopic rows. They are therefore weaker than hard constraints but stronger than unregularized weak directions. The distinction is essential for angular classes in isotope-poor data sets. The global linear

problem is solved in a column-scaled, rank-revealing singular-value basis; the same truncation threshold is used for the correction vector, covariance matrix, numerical rank, condition number, and reported correlation diagnostics. Each fit is then assigned an explicit model status. Models are rejected when they are rank deficient, singular, severely ill conditioned, unsupported by the requested molecule coverage, or dominated by nearly identical class directions. Less severe but chemically informative correlations are reported as review items rather than hidden in the covariance matrix.

Anhydride Test Set

Maleic, phthalic, and succinic anhydrides were selected because they form a compact homologous set for which independent isotopologue-based refinements are available. These data serve two roles. First, ordinary molecule-by-molecule fits using the full isotopologue information provide validation geometries. Second, the same family can be used as an integration set for a target anhydride represented by only one observed species. The deployment question is therefore not whether a single poor-data molecule can determine several classes by itself; it cannot. The question is whether chemically related anhydrides can supply enough complementary information to refine more than one class for such a target.

Table 1: Identifiability of the parent-only anhydride class-correction problem. With one parent molecule, only one effective global correction is chemically admissible; separating short C–C, long C–C, carbonyl C–O, and single C–O would be underdetermined. Pooling three parent molecules makes the four-class model full rank.

Case	Parents	Obs.	Classes	Rank
Maleic parent	1	2	1	1
Phthalic parent	1	2	1	1
Succinic parent	1	2	1	1
Three-parent ensemble	3	6	4	4

Table 2: Class corrections obtained from the parent-only anhydride ensemble. Only heavy-atom stretches are fitted. C–H distances are retained at the BDPCS3 reference values.

Class	Matches	Molecules	Correction / Å	σ / Å
CC_short	14	2	+0.00026	0.00768
CC_long	15	3	+0.00595	0.02226
CO_carbonyl	10	3	-0.02084	0.11862
CO_single	4	3	+0.01836	0.10427

Table 1 is the deployment test. A single parent molecule, or any one-species target, can support only one effective correction if the model is to remain chemically and statistically meaningful. Pooling three anhydrides increases the number of resolvable variables: the four-class heavy-atom model becomes rank complete because each shared correction is sampled through a different topology. This is the practical use case for a new anhydride with only one isotopologue: the target contributes its own parent constants, while the related anhydrides supply the additional class-discriminating information. The fitted C–C corrections are well behaved, whereas carbonyl and single C–O corrections are strongly anticorrelated and have large uncertainties. This is the desired diagnostic behavior: the method increases the number of admissible classes beyond the isolated one-species limit, while still reporting which classes remain weak.

Table 3: Validation of the parent-only anhydride ensemble against ordinary single-molecule refinements using the available isotopologues. Values are class averages over all matched primitive coordinates. Stretch differences are reported in mÅ in the last column.

Class	Unit	n	QC	ensemble	separate	$\Delta_{\text{ens-sep}}$
CC.long	Å / mÅ	7	1.4950	1.5010	1.4959	+5.1
CC_short	Å / mÅ	7	1.3807	1.3810	1.3814	-0.4
CO_carbonyl	Å / mÅ	6	1.1886	1.1678	1.1883	-20.5
CO_single	Å / mÅ	6	1.3855	1.4039	1.3864	+17.5

The validation in Table 3 compares this parent-only deployment against the independent fits that use the full isotopologue information. The short C–C class agrees within 1 mÅ, and the long C–C class within about 5 mÅ. The C–O classes are much less secure, with deviations

of about 20 mÅ and large formal uncertainties. Thus the parent-only ensemble recovers chemically meaningful C–C corrections that no individual parent molecule can determine, while correctly flagging the C–O separation as underconstrained by this three-member set.

Glycine Conformer Test

The anhydride comparison validates transfer across a homologous chemical family and then tests the parent-only deployment. A complementary validation question is whether class definitions can be transferred between different conformers of the same molecule, where the chemical graph is conserved but the geometry, symmetry, atom ordering, and isotopic information are not identical. Glycine conformers I and II provide such a case. The two calculations used the existing semiexperimental benchmark inputs directly; the ensemble layer builds an independent non-redundant GIC model for each conformer and then maps both models to the same global class corrections.

Table 4: Glycine I/II ensemble refinement using two class definitions. The element/value model uses ordinary atom symbols and C–O value windows. The synthon model replaces those coarse atom types by continuous Z_{eff} prototypes with a threshold of 0.035. C–H stretches are retained at the BDPCS3 reference values in both variants.

Class definition	Rank	Scaled condition	WRMS before	WRMS after
Element and value windows	10	589.81	4900.88	30.47
Continuous synthon Z_{eff} types	10	227.15	4900.88	25.96

Both models are rank complete, but the synthon-typed model lowers the weighted RMS from 30.47 to 25.96 and improves the scaled condition number from 589.81 to 227.15. The improvement is modest but important mechanistically: it is obtained without manually identifying corresponding atoms in the two conformers. The carboxyl carbon, methylene carbon, carbonyl oxygen, hydroxyl oxygen, and exchangeable hydrogens are recognized by their continuous local environments, not by atom labels; C–H stretches are kept at the BDPCS3 reference geometry.

Table 5: Threshold scan for the continuous glycine synthon atom types. The model is accepted and unchanged from 0.010 to 0.075, showing that the selected threshold is inside a stable plateau. At 0.100, chemically distinct atom types begin to overlap, the numerical rank drops, and the acceptance policy rejects the singular model.

Z_{eff} threshold	Status	Rank	Cond.	WRMS	Min. matches
0.010	accepted	10	227.15	25.96	3
0.035	accepted	10	227.15	25.96	3
0.075	accepted	10	227.15	25.96	3
0.100	rejected	9	∞	33.09	3
0.150	rejected	9	∞	33.09	3
0.250	rejected	9	∞	33.09	3

Table 6: Continuous synthon classes used for the glycine I/II fit. The prototype column lists the Z_{eff} values defining the class; each primitive coordinate matches when the chemically relevant Z_{eff} signature is within 0.035 of the prototype. Corrections are in internal-coordinate units.

Class	Type	Z_{eff} prototype	Matches	Correction
Ccarb-Cmethylene	stretch	5.91, 5.88	8	-0.00747
Cmethylene-N	stretch	5.88, 6.96	8	-0.00455
Ccarb-Ocarbonyl	stretch	5.91, 7.865	8	-0.00250
Ccarb-Ohydroxyl	stretch	5.91, 7.95	9	-0.00446
N-H	stretch	6.96, 0.868	5	-0.00364
Ohydroxyl-H	stretch	7.95, 0.898	9	0.00817
Ccarb-Cmethylene-Ocarbonyl	bend	5.88, 5.91, 7.865	16	-0.00382
Ccarb-Cmethylene-N	bend	5.91, 5.88, 6.96	3	0.00250
H-Cmethylene-H	bend	0.838, 5.88, 0.838	13	-0.00870
H-N-H	bend	0.868, 6.96, 0.868	16	-0.01974

The glycine result therefore tests a different part of the formalism from the anhydrides. The anhydrides show that soft priors can regularize weak class directions in a sparse homologous set. Glycine shows that transferable classes do not have to be hand-coded as molecule-specific atom lists: continuous synthon types provide a route to chemically meaningful class definitions across conformers.

Table 7: Geometry-level comparison between the synthon-typed glycine I/II ensemble refinement and independent conformer refinements. Values are class averages over matched primitive coordinates. Stretch differences are reported in mÅ in the last column; angle differences are reported in degrees.

Class	Unit	n	QC	ensemble	separate	$\Delta_{\text{ens-sep}}$
Ccarb_Cmethylene_N	deg	2	113.43	113.58	113.68	-0.10
Ccarb_Cmethylene_Ocarbonyl	deg	2	124.19	123.97	123.54	+0.43
H_Cmethylene_H	deg	2	106.51	106.01	106.41	-0.40
H_N_H	deg	2	106.65	105.52	106.34	-0.82
Ccarb_Cmethylene	Å / mÅ	2	1.5234	1.5160	1.5170	-1.1
Ccarb_Ocarbonyl	Å / mÅ	2	1.2037	1.2012	1.2053	-4.1
Ccarb_Ohydroxyl	Å / mÅ	2	1.3456	1.3411	1.3433	-2.1
Cmethylene_N	Å / mÅ	2	1.4554	1.4509	1.4493	+1.6
N_H	Å / mÅ	4	1.0120	1.0083	1.0101	-1.8
Ohydroxyl_H	Å / mÅ	2	0.9728	0.9810	0.9781	+2.8

The geometry comparison confirms that the conformer ensemble is chemically more transferable than the sparse anhydride series. Shared heavy-atom stretch corrections reproduce the independent conformer refinements within a few mÅ, while C–H distances remain fixed at the BDPCS3 reference values. The largest deviations are angular, especially the H–N–H bend, where the common class correction shifts both conformers in the same direction but does not fully reproduce the separate refinements. This is again chemically informative: conformer-transferable heavy-atom classes are well defined by the data, whereas flexible hydrogen angular frames require either richer isotopic coverage or explicit prior information.

Outlook

The present letter skeleton establishes a validation-to-deployment protocol.¹ Anhydrides and glycine provide cases where multiple related molecules and isotopologue-rich refinements allow the transferable class model to be checked against independent semiexperimental geometries. The same machinery can then be applied in two obvious ways. First, a one-species target, for example another anhydride for which only the parent or one isotopologue is known, can be integrated with the validated anhydride family; treated alone it would permit only one

effective correction, whereas the family-integrated model can support several class-specific corrections. Second, multistructure can search the semiexperimental geometry library for molecules most similar to the families detected in the target, recompute those selected references at the same level as the target calculation, and transfer the resulting class corrections to the new molecule. The formalism is also relevant beyond semiexperimental structures, including force-field calibration, systematic correction of quantum-chemical geometries, and transferability tests across molecular families.

References

- (1) Barone, V. MORPHEUS multi-molecule class-correction refinement working implementation. 2026; Working repository benchmark and software manuscript scaffold.